PRODUCT REQUIREMENTS DOCUMENT

BreakfastWise

Breakfast-demand forecasting and food-preparation planning for small and medium-sized hotels

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Important: Market assumptions and success metrics are not verified results. They are labelled for validation before real-world claims are made.

1. Executive Summary

BreakfastWise is a breakfast-demand forecasting dashboard for hotels. It uses expected occupancy and historical breakfast-consumption data to estimate demand, plan food quantities, and compare forecasts with actual consumption and wastage. **Core need (from the MRD):** kitchen teams at small and medium-sized hotels need a practical way to estimate breakfast demand so they can reduce avoidable overproduction and shortages without relying only on manual judgement.

The product covers the journey from data collection → validation → forecasting → comparison → visualization → preparation planning. Segment, persona and targets are MRD assumptions, not validated results.

2. Product Goal

To help hotel kitchen staff make better breakfast-preparation decisions by turning expected occupancy and past breakfast records into a clear, explainable demand forecast.

3. Target Users

Initial segment (ASSUMPTION): independent small and medium-sized hotels serving about 50–200 breakfast guests per day.

Persona	Goal	Pain Point (to validate)	Current Workaround	Role in the System
Primary: Head chef / kitchen manager	Prepare suitable quantities; avoid shortage and excess	Demand varies with occupancy, weekday and groups	Experience, previous counts, paper notes, spreadsheets	Enters expected occupancy; reviews forecast; plans quantities
Secondary: Hotel manager	Track forecast performance, food cost and wastage	No single view of occupancy, attendance and discarded food	PMS reports, spreadsheets, manual summaries	Views dashboard, trends and accuracy
Supporting: Kitchen staff	Record prepared, served and remaining quantities	Manual recording may be inconsistent	Paper registers, verbal updates, logs	Records actuals after service

4. Problem Statement

Hotels often struggle to predict breakfast demand, leading to food wastage or shortages. Breakfast is prepared before the final guest count is known: underestimation causes shortages, and overestimation leaves excess or discarded food. A simple data-driven system is needed to estimate demand from expected occupancy and historical consumption.

Evidence status: this is a starting hypothesis; no verified hotel interviews or waste baseline exist yet.

5. Product Objectives

- Collect expected-occupancy and breakfast-consumption data in a structured way; validate and persist it.
- Forecast breakfast demand with an explainable calculation and flag insufficient data.
- Compare predicted demand with actual consumption and show history and trends.
- Support food-quantity planning and shortage/excess analysis.
- Record basic item-level wastage if confirmed by the team and mentor.

6. MRD Needs Addressed

MRD Need	Product Response
Estimate demand from expected occupancy and prior attendance	Demand forecasting (F3)
Use available hotel data in a simple workflow	Data entry, validation, historical database (F1, F2)
Avoid misleading forecasts	Data-sufficiency warning (F4)
Understand forecast performance; see trends	Forecast vs. actual, history, dashboard (F5–F7)
Support food-quantity planning	Quantity analysis and recommendations (F8, F9)
Link attendance with discarded food	Basic wastage tracking (F10, proposed)

7. Product Scope

In scope: expected occupancy, breakfast consumption, validation, persistence, forecasting, forecast history, actual-vs-predicted comparison, dashboard, planned quantity analysis, shortage/excess identification, recommendations, and basic wastage tracking (pending confirmation).

Out of scope: universal PMS integration, autonomous purchasing, guaranteed savings, automatic model retraining, unverified event-data feeds, and multi-hotel benchmarking (Section 23).

8. Product Capabilities & Priorities

ID	Capability	Priority
F1	Expected-occupancy and breakfast-consumption data entry	MUST
F2	Data validation and database persistence	MUST
F3	Breakfast demand forecasting	MUST
F4	Forecast reliability / insufficient-data indication	SHOULD
F5	Forecast vs. actual comparison	SHOULD
F6	Forecast history and demand trends	SHOULD
F7	Interactive dashboard and date selection	SHOULD
F8	Planned quantity vs. predicted demand analysis	COULD
F9	Food-preparation recommendation	COULD
F10	Basic wastage tracking (proposed MVP; confirm with team/mentor)	COULD

9. MUST-HAVE Requirements

F1 — Data Entry: Staff enter expected occupancy by date (optional room count, notes) and, after service, actual breakfast consumption (optional verified occupancy, notes). *Why Must-Have:* forecasting cannot operate without input data.

F2 — Validation & Persistence: The system validates required fields, dates and non-negative values, and detects missing, invalid, negative and duplicate/conflicting records, then stores valid records in the database. *Why Must-Have:* forecast quality depends on trustworthy history.

F3 — Forecasting: The system uses expected occupancy and historical consumption to estimate breakfast guests (Section 11). *Why Must-Have:* forecasting is the central capability.

10. Data Fields

Field	Description	Required / Status
Date / weekday	Calendar date; weekday derived	Yes
Expected occupancy	Guests expected on the date (forecast input)	Yes
Actual breakfast consumption	Breakfast guests actually served (define guests vs. meals)	Yes (historical)
Actual occupancy; room count; notes	Verified count and extra context	Optional
Planned quantity	Food planned for preparation; units must be defined	Optional
Item prepared / served / remaining / discarded	For quantity estimation and wastage (F10)	Optional; not in current model
Groups, holidays, events, early departures	Possible demand factors	Optional; manual input only
Predicted demand; Forecast status	Calculated output shown on the dashboard	System-generated

Data plan: use real hotel data if permitted; otherwise clearly labelled sample data to test the workflow. Sample data cannot prove real-world accuracy or wastage reduction.

11. Forecasting Logic

The first version uses a simple, transparent calculation that can be explained and verified from stored data.

Step 1: Consumption Rate = Total Breakfast Consumption ÷ Total Occupancy
Step 2: Predicted Demand = Expected Occupancy × Historical Consumption Rate

Example: historical occupancy 1,000 and consumption 800 give a rate of 0.80; expected occupancy of 150 gives 150 × 0.80 = 120 breakfast guests. If total historical occupancy is zero, show an insufficient-data message; predictions are rounded up to a whole guest.

Forecasting layers

- **Historical patterns:** past attendance and occupancy, including weekday/weekend once enough records exist.
- **Upcoming conditions:** expected occupancy plus optional manual context (groups, holidays, events, early departures).
- **Updated prediction:** adjust only when impact is established from past data; no arbitrary adjustments.
- **Learn from actuals:** record actual attendance, compare and re-evaluate on demand; automatic retraining is out of scope.

Model direction: the baseline is compared with Random Forest and Gradient Boosting regressors, and a model is chosen only after time-based testing (train on earlier dates, test on later dates).

12. Forecast Reliability

Before showing a forecast, the system checks how much valid history exists and warns: “Insufficient historical data. Add more records for a more reliable forecast.” Proposed default thresholds (adjustable):

Valid historical records	Forecast status
Fewer than 7	Insufficient data (warning shown)
7 to 29	Low confidence
30 or more	Reliable

13. SHOULD-HAVE Requirements

F4 — Insufficient-Data Indication: warn when valid history is below the Section 12 thresholds.

F5 — Forecast vs. Actual: show predicted demand, actual attendance, difference and forecast error.

F6 — Historical Trends: show forecast history and demand patterns in tables or charts.

F7 — Dashboard: date selection with expected occupancy, predicted demand, actual consumption (when available) and forecast status; sample data is visibly labelled.

14. COULD-HAVE Requirements

F8 — Planned Quantity vs. Predicted Demand: estimate item quantities from predicted guests using portion assumptions (Item Quantity = Predicted Guests × Portion per Guest), then compare with the planned quantity using a proposed ±5% tolerance. Item suggestions are not learned predictions without item-level history.

Status	Condition
Shortage	Planned is more than 5% below the estimate
Balanced	Planned is within ±5% of the estimate
Excess	Planned is more than 5% above the estimate

F9 — Recommendation: a simple action, e.g. “Increase preparation by approximately 10 servings”, “Reduce by approximately 8 servings”, or “Planned quantity is appropriate”.

F10 — Basic Wastage Tracking (proposed MVP, pending confirmation): record per item prepared, served, remaining and actually discarded quantities. Remaining food is not automatically waste, so both are recorded separately.

Wastage % = Quantity Discarded ÷ Quantity Prepared × 100

15. Jira Epic Alignment

Jira Epic	PRD Coverage
Epic 1 — Hotel Data Management	F1, F2; expected occupancy, consumption, validation, persistence
Epic 2 — Breakfast Demand Forecasting	F3, F4, F5, F6; baseline and model evaluation, data sufficiency, comparison
Epic 3 — Forecast Dashboard	F6, F7; summaries, trends, date selection, accuracy view
Epic 4 — Food Preparation Planning	F8, F9, F10; item quantities, shortage/excess, recommendations, wastage

16. Jira Story-Level Relationship

Jira stories are the implementation work under these capabilities. The PRD defines what the product must provide; Jira epics and stories define how the team builds and tracks it.

17. Success Metrics

Proposed targets from the MRD, not achieved results.

Metric	Proposed target	How to evaluate
Guest-demand forecasting	MAE ≤ 10 guests/day	Held-out dates, actual vs. predicted
Baseline comparison	Better MAE than historical-average baseline	Same time-based test set
Item portion estimation	Within ±15% of actual demand	Only with reliable item-level actuals
Wastage reduction	≥ 15% in a controlled pilot	Defined baseline, consistent units
Data validation	≥ 95% of invalid test records flagged	Missing, negative, invalid date/number
Response time; workflow tests	Under 3 s locally; 100% of test cases pass	Repeated local runs; agreed test suite

- **Forecast Error** = Actual − Predicted; **MAE** = (1/n) × Σ |Actual − Predicted| (n = compared days).
- **MAPE (optional)** = (1/n) × Σ (|Actual − Predicted| ÷ Actual) × 100, excluding zero-actual days.
- **Validation Accuracy** = invalid records flagged ÷ total invalid test records × 100.

18. Feature-to-MRD Traceability

Feature	MRD Need
F1 Data Entry	Reliable collection of occupancy and attendance data
F2 Validation & Persistence	Trustworthy historical data
F3 Forecasting	Estimate demand and reduce uncertainty
F4 Data Warning	Prevent unreliable interpretation
F5 Forecast vs. Actual	Understand forecast performance
F6 Historical Trends	Identify demand patterns
F7 Dashboard	Single view for kitchen and hotel manager
F8 Quantity Analysis	Support food-quantity planning
F9 Recommendation	Support preparation decisions
F10 Wastage Tracking	Link attendance with discarded food

19. Key Risks

Risk	Impact	Mitigation
Insufficient history	Unreliable forecast	Show warning; sample data for workflow tests only
Incorrect staff-entered data	Lower forecast quality	Validate all important fields
Inconsistent definitions (guests vs. meals, units)	Misleading comparisons	Define units and counts first
Unusual occupancy or events	Higher forecast error	Compare with actuals; manual context inputs
No expected occupancy for the date	Forecast cannot be calculated	Prompt user to enter it
Missing planned quantity	Gap cannot be calculated	Clearly show missing input

20. User Flows

Flow 1 — Data Entry: Login → Data Management → Select Date → Enter Expected Occupancy / Consumption → Validate → Correct if invalid → Save → Confirmation.

Flow 2 — Forecast: Open Forecast → Select Date → Retrieve history → Check data sufficiency → Calculate rate → Calculate predicted demand → Display forecast.

Flow 3 — Dashboard: Open Dashboard → Select date → View summary → Review trend → Compare actual when available.

Flow 4 — Preparation Planning: View predicted demand → Enter planned quantity → Compare → Shortage/Balanced/Excess → Recommendation.

Flow 5 — Five-minute demo (MRD): Select date → Enter expected occupancy (optional group context) → Review forecast or insufficient-data notice → Plan preparation from portion assumptions → Record actuals (and prepared, served, remaining, discarded if approved) → Compare forecast vs. actual and trends, with sample data labelled.

21. Functional Requirements

Area	Requirements
Data Management	FR1: Enter expected occupancy. FR2: Enter actual consumption. FR3: Validate fields, dates, non-negative numbers. FR4: Save valid records. FR5: Retrieve history.
Forecasting	FR6: Calculate consumption rate. FR7: Generate predicted demand. FR8: Identify insufficient data. FR9: Use weekday/manual context only when supported by data.
Analysis	FR10: Compare predicted and actual. FR11: Calculate forecast error and MAE. FR12: Show trends.
Dashboard	FR13: Select dates. FR14: Show summaries. FR15: Charts/tables. FR16: Label sample data.
Preparation Planning	FR17: Estimate item quantities. FR18: Compare planned and estimated. FR19: Identify shortage/excess. FR20: Recommend action.
Wastage (proposed)	FR21: Record prepared, served, remaining, discarded per item. FR22: Calculate wastage %.

22. Non-Functional Requirements

- **Usability:** simple for kitchen staff with basic computer knowledge.
- **Performance:** actions respond within about 3 seconds locally.
- **Reliability:** saved records remain after application restart.
- **Data Integrity:** invalid records are not stored as valid.
- **Security:** only authorized users may add or modify hotel data.

23. Explicit Out-of-Scope Items

Out-of-Scope Item	Reason
Integration with every PMS	Needs vendor-specific access; validate one data pathway first
Autonomous purchasing / supplier ordering	Not needed to show forecasting and planning
Guaranteed savings or waste reduction	Must be measured in a controlled pilot
Continuous/automatic model retraining	Defer until reliable real data and safe deployment exist
Unverified event-data feeds	Use manual inputs or omit
Multi-hotel benchmarking	Needs consent and standardised data

24. Assumptions

- **Segment fit:** hotels with 50–200 breakfast guests/day have a real planning problem and may use a lightweight dashboard.
- **Data access:** hotels can provide consistent occupancy and attendance history.
- **Wastage measurement:** staff can consistently record prepared, served, remaining and discarded quantities.
- Consumption relates usefully to occupancy; quantities use a consistent unit; zero-occupancy days are handled safely.

Unknowns to validate

- Whether the head chef or kitchen manager owns both forecasting and end-of-service recording.
- Whether item-level consumption and portion data exist; if not, start with guest-count forecasting.
- Whether group bookings, holidays, events and early departures are accessible and useful.

25. MVP Definition

MVP	Capability
MVP 1	Expected-occupancy and breakfast-consumption data entry
MVP 2	Validation and database persistence
MVP 3	Breakfast demand forecasting

A successful MVP demonstrates **DATA → FORECAST**; the rest extends toward **ANALYSIS → DECISION**. Wastage tracking (F10) joins the MVP only if the team and mentor confirm it.

26. Acceptance Criteria

1. Staff can enter expected occupancy and breakfast-consumption records.
2. Invalid data is rejected or clearly flagged.
3. Valid records are stored and retrievable from the database.
4. A forecast can be generated from available history.
5. The forecast calculation can be explained during the demo.
6. Insufficient historical data is clearly indicated.
7. Predicted demand can be compared with actual consumption.
8. Forecast history and trends can be viewed on the dashboard.
9. Planned quantity can be compared with predicted demand; shortage/excess and a recommendation are shown.
10. Sample data is visibly labelled; if F10 is confirmed, wastage % is calculated.

27. Demo Scenario

Enter Data → Validate → Save → Select Forecast Date → Enter Expected Occupancy → Calculate Forecast → Show Calculation → Plan Quantities → Record Actuals → Compare Actuals → View Trend → Identify Shortage/Excess → (If confirmed) Show Wastage % → Show Recommendation.

28. Requirements Traceability Summary

PRD Capability	Priority	Demo Evidence
F1 Data Entry	MUST	New occupancy/consumption record
F2 Validation & Persistence	MUST	Invalid input rejected + database record
F3 Forecasting	MUST	Forecast result + calculation
F4 Data Sufficiency	SHOULD	Warning for limited history
F5 Forecast vs. Actual	SHOULD	Comparison result and error
F6 Trends	SHOULD	Historical chart/table
F7 Dashboard	SHOULD	Date-based dashboard
F8 Quantity Analysis	COULD	Shortage/excess calculation
F9 Recommendation	COULD	Preparation action message
F10 Wastage Tracking	COULD	Wastage % for a recorded day (if confirmed)

29. Technology Stack

Layer	Technology (from the MRD technical context)
Backend	FastAPI application with occupancy, consumption and history routers
Database	PostgreSQL with psycopg
Validation	Pydantic models (date and non-negative numeric checks)
Frontend	Static assets served from the application's static directory
Testing	Pytest and HTTPX
Forecasting / ML	scikit-learn not yet in requirements.txt; Random Forest and Gradient Boosting are candidates; code not yet verified

30. Future Enhancements

- Choose a Random Forest or Gradient Boosting model after time-based evaluation against the baseline.
- Scheduled retraining once reliable real data and a safe validation process exist.
- Verified holiday and event data sources instead of manual inputs.
- Integration with one hotel PMS data pathway; alerts when predicted demand differs sharply from plan.

31. Final Product Statement

BreakfastWise is a focused hotel decision-support system that turns expected occupancy and breakfast records into an understandable demand forecast and practical preparation guidance. Market fit, data access, item-level forecasting and waste-reduction impact remain to be validated. **Next steps:** review with team and mentor, collect hotel-staff evidence, confirm MVP scope (including wastage), and measure performance on defined test data.

DATA → FORECAST → ANALYSIS → DECISION